

2015

Year 12 Mathematics

Trial Examination

Teacher Setting Paper: Mrs T Finch

Head of Department: Mrs M Hill

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
- Board approved calculator may be used
- Write your answers to Section 1 on the multiple choice answer sheet provided
- Write your answers to Section 2 in the answer booklets provided. Start a new booklet for each question
- Write your student number only on the front of each booklet
- A table of standard integrals is provided at the back of this paper
- In Questions 11-16, show relevant mathematical reasoning and/or calculations

Total marks – 100

Section 1 Pages 2-4

10 marks

- Attempt all questions 1-10
- Allow 15 minutes for this section

Section 2 Pages 5 - 12

90 marks

- Attempt all questions 11-16
- Allow 2 hour and 45 minutes for this section

This examination paper does not necessarily reflect the content or format of the Higher School Certificate Examination in this subject

Section 1 (10 marks)

Attempt Questions 1 -10

Allow 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1 -10

1. What is $\sqrt[3]{\frac{2.7-8.1 \times 0.04}{3.4 \times 10^{-2}}}$ correct to 3 significant figures?

- (A) 4.11
- (B) 4.12
- (C) 4.118
- (D) 4.119

2. If $3\sqrt{5} + \sqrt{20} = \sqrt{a}$, then $a =$?

- (A) 5
- (B) 25
- (C) 125
- (D) 15

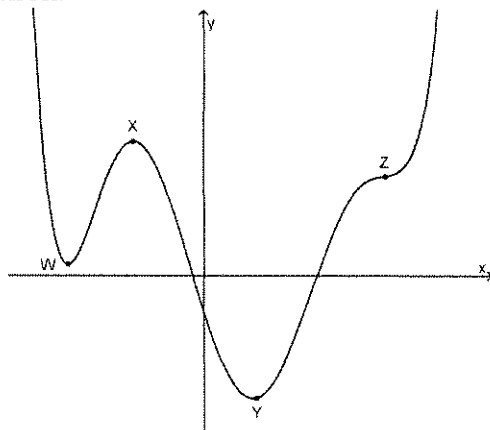
3. What is the exact value of $\cos \frac{7\pi}{6}$?

- (A) $\frac{\sqrt{3}}{2}$
- (B) $-\frac{\sqrt{3}}{2}$
- (C) $\frac{1}{\sqrt{2}}$
- (D) $-\frac{1}{\sqrt{2}}$

4. $\int_1^4 f(x) dx = 2$ then $\int_1^4 (2f(x)+3) dx$ is equal to

- (A) 2
- (B) 13
- (C) 7
- (D) 10

5. The diagram below shows the graph of $y = f(x)$. Points W , X , Y and Z are stationary points and Z is a point of inflexion.



Which of the points on $y = f(x)$ corresponds to the description

$$y > 0 \quad \frac{dy}{dx} = 0 \quad \frac{d^2y}{dx^2} > 0$$

- (A) W
 (B) X
 (C) Y
 (D) Z
6. $\lim_{\theta \rightarrow 0} \frac{\sin \theta}{4\theta} =$
- (A) 1
 (B) 4
 (C) $\frac{1}{4}$
 (D) 0

7. The function $f(x) = -3 \cos\left(\frac{\pi x}{5}\right)$ has a period of

- (A) $\frac{\pi}{5}$
 (B) $\frac{\pi}{10}$
 (C) 3
 (D) 10

8. The limiting sum of the series $1 - 2p + 4p^2 \dots$ is $\frac{4}{7}$, find the value of p

- (A) $\frac{3}{8}$
- (B) $-\frac{3}{8}$
- (C) $\frac{2}{7}$
- (D) $-\frac{2}{7}$

9. What are the solutions to $3x^2 - 7x - 3 = 0$

- (A) $x = \frac{7 \pm \sqrt{85}}{6}$
- (B) $x = \frac{-7 \pm \sqrt{85}}{6}$
- (C) $x = \frac{7 \pm \sqrt{13}}{6}$
- (D) $x = \frac{-7 \pm \sqrt{13}}{6}$

10. Which of the following represents the domain of the function: $f(x) = \sqrt{9 - x^2} + \frac{1}{x - 3}$

- (A) $x \neq 3$
- (B) $x < 3$
- (C) $-3 < x < 3$
- (D) $-3 \leq x < 3$

*End of multiple choice section
Section 2 is on the next page*

Section 2 (90 marks)

Attempt Questions 11-16

Allow about 2 hour and 45 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

In Questions 11-16, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use a SEPARATE writing booklet

a) Solve for x : $\frac{6}{x} = x - 5$ 2

b) Solve $|3x - 4| \leq 8$ 2

c) If $\tan \theta = \frac{7}{9}$ and $\cos \theta < 0$, find the exact value of $\operatorname{cosec} \theta$ 2

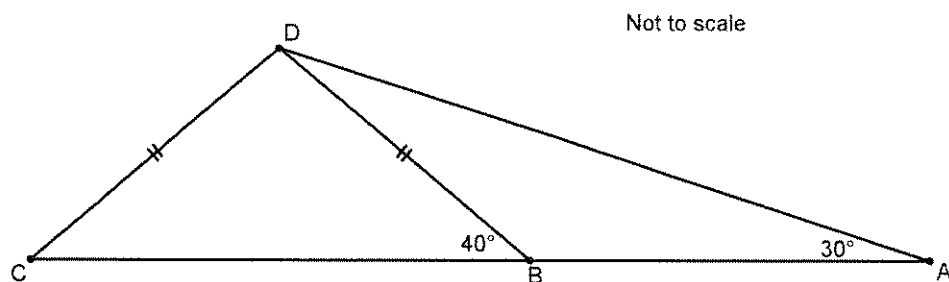
d) Solve for x : $9^{2x-3} = 27$ 2

e) Differentiate with respect to x :

(i) $y = \cos(x^2 - 1)$ 2

(ii) $y = \frac{1 - e^{2x}}{x^3}$ 2

f) In the diagram below ABC is a straight line. If $\angle DBC = 40^\circ$, $\angle DAB = 30^\circ$, $BD = CD$ and $AB = 10 \text{ cm}$, what is the value of BC , to the nearest cm ? 3

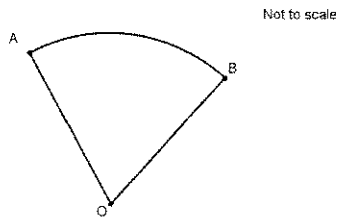


End of Question 11

Question 12 (15 marks) Use a SEPARATE writing booklet

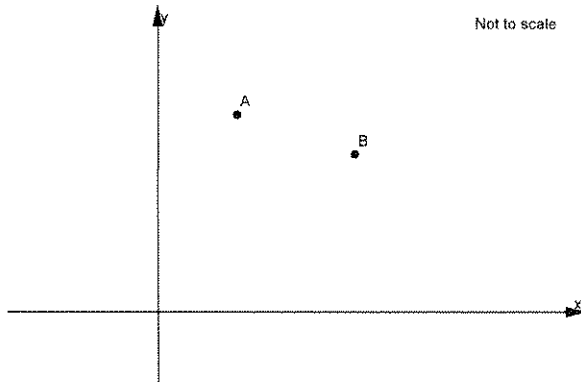
- a) In the diagram AB is an arc of a circle with the centre O . The length of the arc AB is

$\frac{5\pi}{2} \text{ cm}$. The area of the sector AOB is $4\pi \text{ cm}^2$. Find the radius of the sector.



2

- b) The diagram shows the points $A(2,5)$ and $B(5,4)$



- (i) Show that the equation of the line AB is $x + 3y - 17 = 0$ 2
- (ii) Find the coordinates of M the midpoint of AB . 1
- (iii) Show that the equation of the perpendicular bisector of AB is $3x - y - 6 = 0$ 1
- (iv) The perpendicular bisector of AB cuts the x -axis at C . Find the coordinates of C . 1
- (v) Find the area of $\triangle ABC$ 2

- c) Give the exact value of $\log_3\left(\frac{1}{\sqrt{3}}\right)$ 1

- d) Find $\int \sin \frac{x}{2} dx$ 2

- e) Find the equation of the normal to $y = (x^3 - 1)^2$ at the point $x = -1$ 3

End of Question 12

Question 13 (15 marks) Use a SEPARATE writing booklet

a) Consider the function $f(x) = 2x^3 - 3x^2 - 36x + 26$

- (i) Find the co-ordinates of the stationary points of the curve $y = f(x)$ and determine their nature. 3
- (ii) Find the co-ordinates of any point of inflexion. 1
- (iii) Hence sketch the graph of $f(x) = 2x^3 - 3x^2 - 36x + 26$ by showing the above information. 2
- (iv) For what values of x is the curve concave down and decreasing? 1

b) There are three green lolly frogs and one red lolly frog in a jar. One frog is selected at random, eaten, and then a second frog is selected at random and also eaten. Find the probability that:

- (i) The two frogs eaten are both green. 1
- (ii) The red frog is the second one eaten. 1

c) Let A be the point $(-2, 0)$ and $B(6, 0)$. The point $P(x, y)$ is such that $AP \perp PB$.

- (i) Find the gradient of PA . 1
- (ii) Hence, find the equation of the locus of P . 2

d) If α and β are the roots of the quadratic equation $3x^2 + 8x + 7 = 0$, find the value of

- (i) $\alpha + \beta$ 1
- (ii) $\alpha\beta$ 1
- (iii) $\alpha^2 + \beta^2$ 1

End of Question 13

Question 14 (15 marks) Use a SEPARATE writing booklet

a) Consider the parabola $2y = x^2 - 4x$

(i) Find the coordinates of the focus

2

(ii) Find the equation of the directrix.

1

(iii) What is the length of the latus rectum?

1

b) Solve: $2\sin\theta + \sqrt{3} = 0$ for $0 \leq \theta \leq 2\pi$

2

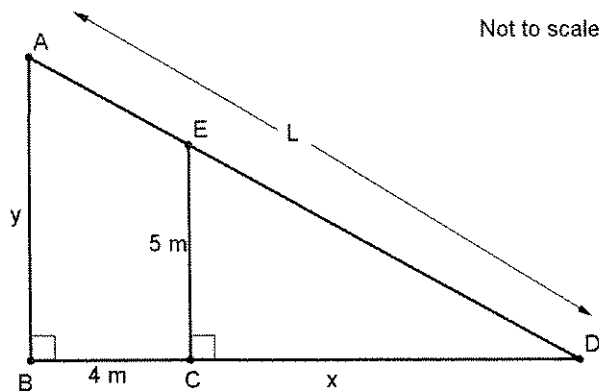
c) Evaluate: $\int_1^3 (6e^{3x} + 1) dx$

2

d) A sum of \$2 000 is deposited at the start of each year in an account that earns 10% interest per annum. Find the total value of the investment at the end of the 15th year, correct to the nearest dollar

2

e) A 5 metre high fence stands 4 metres from the wall of a house. A farmer wishes to reach a point, A, on the wall by the use of a ladder, L, that can reach from the ground outside the fence to the wall as shown in the diagram below.



(i) Prove that $y = \frac{5(x+4)}{x}$

1

(ii) Hence show that $L^2 = (x+4)^2 \left(1 + \frac{25}{x^2}\right)$

2

(iii) Hence find the length of the shortest ladder that can reach from the ground outside the fence to the wall. (Answer to 1 decimal place)

2

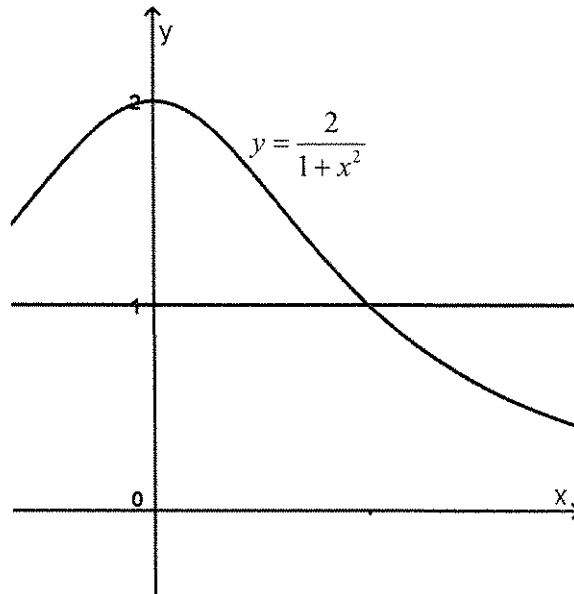
End of Question 14

Question 15 (15 marks) Use a SEPARATE writing booklet

- a) A particle moves along a smooth horizontal surface according to $x = 10 + 8t - 2t^2$, with x in metres and t in seconds.

- | | | |
|-------|--|---|
| (i) | Find when the particle is at the origin. | 1 |
| (ii) | Find where the particle is stationary. | 2 |
| (iii) | Show that the particle has constant acceleration. | 1 |
| (iv) | How far has the particle travelled in the first 6 seconds. | 2 |

- b) The region between the curve $y = \frac{2}{1+x^2}$, the y -axis and the line $y = 1$ shown below, is rotated about the y -axis.



- (i) Show that the volume of the solid formed is given by

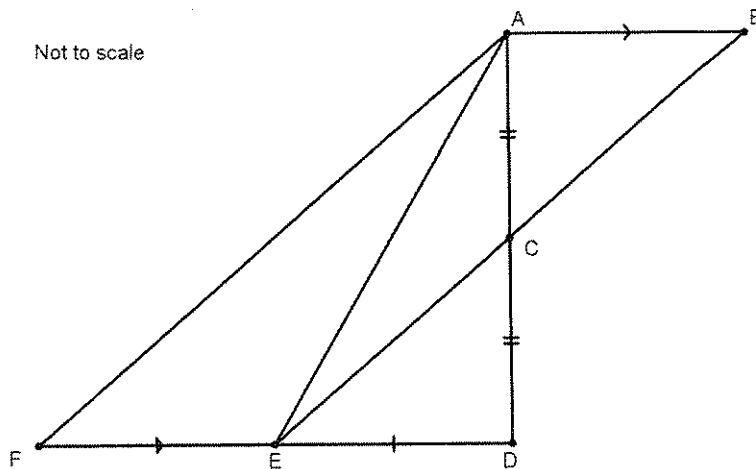
$$V = \pi \int_1^2 \left(\frac{2}{y} - 1 \right) dy \quad 1$$

- (ii) Hence find its volume. 2

Question 15 is continued on page 10

Question 15 (continued)

- c) In the diagram $AB \parallel FD$. $\triangle ADF$ is right angled, C is the midpoint of AD and E is the midpoint of FD .



Copy or trace this diagram into your answer booklet.

- | | | |
|-------|--|---|
| (i) | Explain why $\angle CED = \angle ABC$ | 1 |
| (ii) | Show that $\triangle CDE \equiv \triangle CAB$ | 2 |
| (iii) | Show that $AF = 2BC$ | 1 |

- d) Consider the function $y = \log_e(x^2 + 1)$. Using Simpson's Rule with 5 function values to approximate $\int_0^4 \log_e(x^2 + 1) dx$

2

End of Question 15

Question 16 (15 marks) Use a SEPARATE writing booklet

- a) The population of a species of bacteria P , at time t minutes, grows such that

$$P = 2000e^{kt} \text{ where } k \text{ is a positive constant.}$$

- (i) Show that the rate of increase of the population is proportional to the size of the population at that time, that is $\frac{dP}{dt} = kP$. 1
- (ii) Given that the initial population doubles after 4 minutes, calculate the value of k correct to 3 significant figures. 2
- (iii) Find the population after 6 minutes, correct to the nearest whole number. 2

- b) Henry borrows \$200 000 which is to be repaid in equal monthly instalments of \$ M . The interest rate is 7.2% per annum reducible, calculated monthly.

- (i) Show that the amount \$ A_n , owing after the n th month is given by the formula: 2
- $$A_n = 200\,000 \times 1.006^n - M(1 + 1.006 + 1.006^2 + \dots + 1.006^{n-1})$$
- (ii) The minimum monthly repayment is the amount required to repay the loan in 25 years. Find the minimum monthly repayment, \$ M . 3

Question 16 is continued on page 12

Question 16 (continued)

- c) On a factory production line a tap opens and closes to fill empty containers with liquid. As the tap opens, the rate of flow of liquid, R , litres per second, increases for the first 10 seconds according to

$$R = \frac{6t}{50}$$

After 10 seconds, the rate of flow remains constant until the tap begins to close.

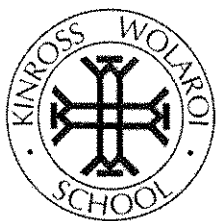
As the tap closes, the rate of flow decreases at a constant rate for 10 seconds, after which the tap has fully closed and the rate of flow is zero.

- (i) Show that, while the tap is fully open, the volume in the container, in litres, at any time t , is given by

$$V = \frac{6}{5}(t - 5) \quad 2$$

- (ii) For how many seconds must the tap remain fully open in order to exactly fill a 120 litre container with no spillage? 3

End of Examination



2015

Number: _____

Year 12 Mathematics

Trial Examination: Multiple Choice Answer Sheet

For multiple choice questions, choose the best answer A, B, C or D and fill in the correct circle.

1. ☐ A ☐ B ☐ C ☐ D
2. ☐ A ☐ B ☐ C ☐ D
3. ☐ A ☐ B ☐ C ☐ D
4. ☐ A ☐ B ☐ C ☐ D
5. ☐ A ☐ B ☐ C ☐ D
6. ☐ A ☐ B ☐ C ☐ D
7. ☐ A ☐ B ☐ C ☐ D
8. ☐ A ☐ B ☐ C ☐ D
9. ☐ A ☐ B ☐ C ☐ D
10. ☐ A ☐ B ☐ C ☐ D

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STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right) \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, \quad x > 0$

2015 Maths Trial
Multiple Choice

- | | |
|-----|------|
| 1 B | 6 C |
| 2 C | 7 D |
| 3 B | 8 A |
| 4 B | 9 A |
| 5 A | 10 D |

① $4.1189 \approx 4.12$

② $3\sqrt{5} + \sqrt{20}$
 $= 3\sqrt{5} + 2\sqrt{5}$
 $= 5\sqrt{5}$

$\sqrt{a} = \sqrt{125}$

③ $\cos \frac{7\pi}{6}$ $\frac{S/A}{\sqrt{1/c}}$
 $= -\cos \frac{\pi}{6}$
 $= -\frac{\sqrt{3}}{2}$



④ $\int_1^4 2f(x) + 3 dx$
 $= 2 \int_1^4 f(x) dx + [3x]_1^4$
 $= 2 \times 2 + 12 - 3$
 $= 13$

⑤ $y > 0$ $y' = 0$ $y'' > 0$

⑩ $f(x) = \sqrt{9-x^2} + \frac{1}{x-3}$

Domain $-3 \leq x < 3$

⑥ $\lim_{\theta \rightarrow 0} \frac{\sin \theta}{4\theta} = \frac{1}{4} \lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta}$
 $= \frac{1}{4}$

⑦ $f(x) = -3 \cos \left(\frac{\pi x}{5} \right)$
 Period $= \frac{2\pi}{n}$ $n = \frac{\pi}{5}$
 $= \frac{2\pi}{(\pi/5)}$
 $= 2\pi \times \frac{5}{\pi}$
 $= 10$

⑧ $S_{\infty} = \frac{4}{7}$ $a = 1$ $r = -\frac{2}{7}$
 $\frac{a}{1-r} = \frac{4}{7}$
 $\frac{1}{1-2p} = \frac{4}{7}$
 $4(1+2p) = 7$
 $4+8p = 7$
 $8p = 3$
 $p = \frac{3}{8}$

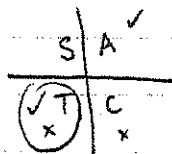
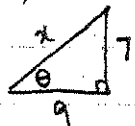
⑨ $3x^2 - 7x - 3 = 0$
 $x = \frac{-(-7) \pm \sqrt{(-7)^2 - 4 \times 3 \times -3}}{2 \times 3}$
 $= \frac{7 \pm \sqrt{85}}{6}$

Q.11

a) $\frac{6}{x} = x - 5$
 $6 = x^2 - 5x$
 $x^2 - 5x - 6 = 0$ ✓
 $(x-6)(x+1) = 0$
 $x = 6, -1$ ✓

b) $|3x-4| \leq 8$
 $-8 \leq 3x-4 \leq 8$ ✓
 $-4 \leq 3x \leq 12$
 $-\frac{4}{3} \leq x \leq 4$ ✓

c) $\tan \theta = \frac{7}{9}$ $\cos \theta < 0$



$x^2 = 7^2 + 9^2$

$x = \sqrt{130}$ ✓

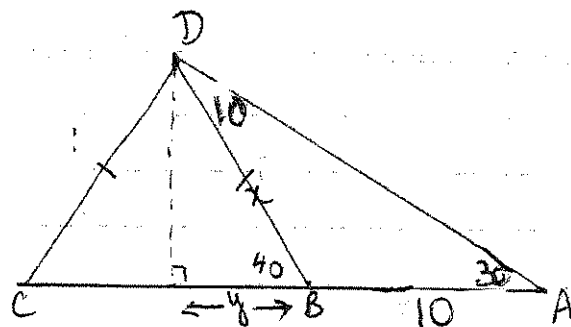
$\operatorname{cosec} \theta = -\frac{\sqrt{130}}{7}$ ✓

d) $9^{2x-3} = 27$
 $(3^2)^{2x-3} = 3^3$ ✓
 $4x-6 = 3$
 $4x = 9$
 $x = \frac{9}{4}$ ✓

e) i) $y = \cos(x^2-1)$
 $y' = -2x \sin(x^2-1)$ ✓

ii) $y = \frac{1-e^{2x}}{x^3}$

$y' = \frac{x^3 \cdot -2e^{2x} - 3x^2(1-e^{2x})}{x^6}$ ✓
 $= \frac{-2xe^{2x} - 3 + 3e^{2x}}{x^4}$



$\frac{x}{\sin 30} = \frac{10}{\sin 10}$ ✓

$x = \frac{10 \sin 30}{\sin 10} \approx 28.79385$

$\cos 40 = \frac{y}{x}$ ✓

$y = \frac{10 \sin 30}{\sin 10} \times \cos 40$

$= 22.05737$

$\approx 22 \text{ cm}$ ✓

$\therefore BC \approx 44 \text{ cm}$

OR Use cosine rule $\angle CDB = \frac{180-80}{100}$

$BC^2 = x^2 + x^2 - 2x \times x \cos \angle CDB$

$BC^2 = 2 \times \left(\frac{10 \sin 30}{\sin 10}\right)^2 - 2 \times \left(\frac{10 \sin 30}{\sin 10}\right)^2 \cos 100^\circ$

$BC \approx 44 \text{ cm}$

OR $\angle DCB = \angle DBC = 40$
 $\angle CDB = 180 - 2 \times 40 = 100$

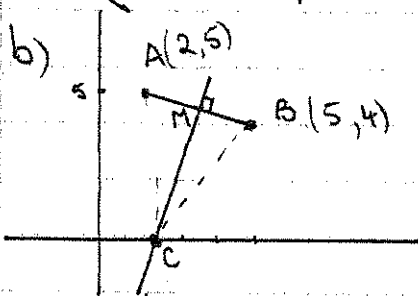
$\frac{BC}{\sin 100} = \frac{x}{\sin 40}$
 $BC = \frac{28.79 \times \sin 100}{\sin 40}$

Q 12

a) $l = r\theta^\circ$ $A = \frac{1}{2} r^2 \theta^\circ$
 $\frac{5\pi}{2} = r\theta$ $4\pi = \frac{1}{2} r^2 \theta$
 $\frac{r^2 \theta}{r\theta} = \frac{8\pi}{(\frac{5\pi}{2})}$ $\therefore r^2 \theta = 8\pi$

$r = \frac{16}{5}$ ✓ $(\frac{31}{2})$

$\left(\frac{16}{5} \times \theta = \frac{5\pi}{2} \right)$
 $\theta^\circ = \frac{25\pi}{32}$



i) $m = \frac{4-5}{5-2} = -\frac{1}{3}$ ✓

$y-5 = -\frac{1}{3}(x-2)$ ✓

$3y-15 = -x+2$

$x+3y-17=0$

ii) $M\left(\frac{2+5}{2}, \frac{5+4}{2}\right) = \left(\frac{7}{2}, \frac{9}{2}\right)$ ✓

iii) $m_{\perp} = 3$ ✓

$y - \frac{9}{2} = 3\left(x - \frac{7}{2}\right)$

$2y-9 = 6x-21$

$6x-2y-12=0$

$3x-y-6=0$

iv) on x-axis $y=0$

$\therefore 3x-6=0$

$x=2$ ✓

$C(2,0)$

b) (v) $d_{AB} = \sqrt{(2-5)^2 + (5-4)^2}$

$= \sqrt{9+1}$

$= \sqrt{10}$ ✓

$\perp d_{MC} = \frac{|1 \times 2 + 3 \times 0 - 17|}{\sqrt{1^2 + 3^2}}$ or use d_{MC}

$= \frac{|-15|}{\sqrt{10}}$

Area $= \frac{1}{2} \sqrt{10} \times \frac{15}{\sqrt{10}}$ ✓

$= 7.5 u^2$

c) $\log_3 \left(\frac{1}{\sqrt{3}}\right) = \log_3 3^{-1/2}$

$= -\frac{1}{2}$ ✓

d) $\int \sin \frac{x}{2} dx = -2 \cos \frac{x}{2} + C$ ✓

e) $y = (x^3 - 1)^2$ at $x = -1$

$y' = 2(x^3 - 1) \times 3x^2$

$= 2((-1)^3 - 1) \times 3 \times (-1)^2$

$= 2 \times -2 \times 3$ ✓

$= -12$

$m_{\perp} = \frac{1}{12}$

$y = ((-1)^3 - 1)^2$
 $= 4$ ✓

$y - 4 = \frac{1}{12}(x - (-1))$ ✓

$12y - 48 = x + 1$

$x - 12y + 49 = 0$ ✓

Q13

a) $f(x) = 2x^3 - 3x^2 - 36x + 26$

i) $f'(x) = 6x^2 - 6x - 36$

$f'(x) = 0$

$x^2 - x - 6 = 0$

$(x-3)(x+2) = 0$ ✓

$x = 3 \quad x = -2$

$f(3) = -55 \quad f(-2) = 70$ ✓

$f''(x) = 12x - 6$

$f''(3) = 36 - 6 = 30 > 0$ ✓

$f''(-2) = 24 - 6 = 18 > 0$ ✓

∴ (3, -55) is a Min. ✓

∴ (-2, 70) is a Max ✓

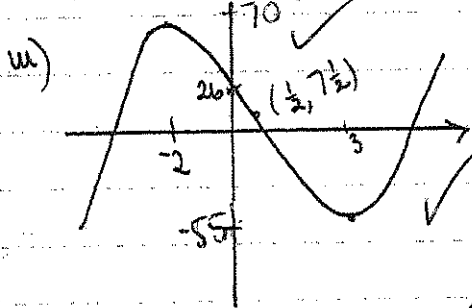
(ii) $f''(x) = 0$

$12x - 6 = 0$

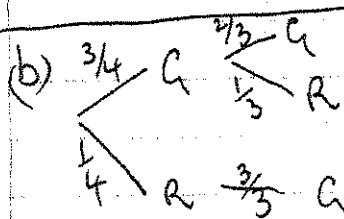
$x = \frac{1}{2}$

$f(\frac{1}{2}) = 2 \times (\frac{1}{2})^3 - 3 \times (\frac{1}{2})^2 - 36(\frac{1}{2}) + 26$
 $= 7\frac{1}{2}$

Inflection at $(\frac{1}{2}, 7\frac{1}{2})$ ✓



iv) $-2 < x < \frac{1}{2}$ ✓



(i) $P(GG) = \frac{3}{4} \times \frac{2}{3} = \frac{1}{2}$ ✓

(ii) $P(GR) = \frac{3}{4} \times \frac{1}{3} = \frac{1}{4}$ ✓

c) A(-2, 0) B(6, 0) P(x, y)

i) $m_{PA} = \frac{y-0}{x-(-2)} = \frac{y}{x+2}$ ✓

ii) $m_{PA} \times m_{PB} = -1$

$\frac{y}{x+2} \times \frac{y}{x-6} = -1$ ✓

$y^2 = -(x+2)(x-6)$

$y^2 = -x^2 - 2x + 6x + 12$

$x^2 - 4x + y^2 = 12$ ✓

or $(x-2)^2 + y^2 = 16$

d) $3x^2 + 8x + 7 = 0$

i) $\alpha + \beta = -\frac{b}{a}$

$= -\frac{8}{3}$ ✓

ii) $\alpha\beta = \frac{c}{a}$

$= \frac{7}{3}$ ✓

iii) $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$

$= \left(-\frac{8}{3}\right)^2 - 2 \times \frac{7}{3}$

$= \frac{64}{9} - \frac{14}{3}$ ✓

$= 2\frac{4}{9}$ ✓

$$14a) 2y = x^2 - 4x$$

$$i) x^2 - 4x + 4 = 2y + 4$$

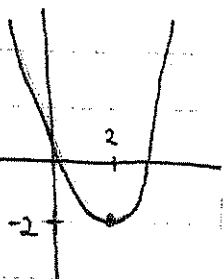
$$(x-2) = 2(y+2)$$

$$V(2, -2)$$

$$4a = 2$$

$$a = \frac{1}{2} \checkmark$$

$$\text{Focus}(2, -\frac{1}{2}) \checkmark$$



$$ii) y = -2\frac{1}{2} \checkmark$$

$$iii) 4a = 2 \checkmark$$

$$b) 2\sin\theta + \sqrt{3} = 0 \quad 0 \leq \theta \leq 2\pi$$

$$\sin\theta = -\frac{\sqrt{3}}{2} \checkmark$$

$$\frac{S}{T} \frac{A}{C} \frac{\pi/3}{\pi/3} \checkmark$$

$$\theta = \pi + \frac{\pi}{3}, 2\pi - \frac{\pi}{3}$$

$$= \frac{4\pi}{3}, \frac{5\pi}{3} \checkmark$$

$$c) \int_1^3 (6e^{3x} + 1) dx$$

$$= \left[\frac{6e^{3x}}{3} + x \right]_1^3 \checkmark$$

$$= [2e^9 + 3] - [2e^3 + 1]$$

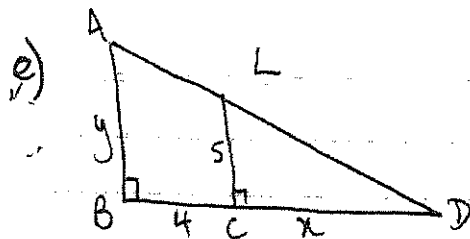
$$= 2e^9 - 2e^3 + 2 \checkmark$$

$$d) 2000 \times 1.1^{15} + 2000 \times 1.1^{14} + \dots + 2000 \times 1.1$$

$$= 2000 [1.1 + 1.1^2 + \dots + 1.1^{15}]$$

$$= 2000 \times \frac{1.1(1.1^{15} - 1)}{1.1 - 1}$$

$$\approx \$69899 \checkmark$$



i) $\triangle ABD \parallel \triangle LCD$ equiangular
 $\therefore$ corresponding sides in same ratio

$$\frac{y}{5} = \frac{x+4}{x}$$

$$y = \frac{5(x+4)}{x}$$

$$ii) AD^2 = AB^2 + BD^2$$

$$L^2 = y^2 + (x+4)^2$$

$$= \frac{25(x+4)^2}{x^2} + (x+4)^2$$

$$= (x+4)^2 \left[1 + \frac{25}{x^2} \right]$$

$$iii) L = (x+4) \left(1 + \frac{25}{x^2} \right)^{1/2}$$

$$\frac{dL}{dx} = (x+4) \times \frac{1}{2} (1 + \frac{25}{x^2})^{-1/2} \times -50x^{-3}$$

$$+ (1 + \frac{25}{x^2})^{1/2} \times 1$$

$$0 = \frac{-25(x+4)}{x^3 \sqrt{1 + \frac{25}{x^2}}} + (1 + \frac{25}{x^2})^{1/2} \times \frac{1}{x^3 (1 + \frac{25}{x^2})^{1/2}}$$

$$0 = -25x - 100 + x^3 \left(1 + \frac{25}{x^2} \right)$$

$$0 = -25x - 100 + x^3 + 25x$$

$$x^3 = 100$$

$$x = \sqrt[3]{100}$$

$$L = (\sqrt[3]{100} + 4) \left(1 + \frac{25}{(\sqrt[3]{100})^2} \right)^{1/2}$$

$$= 12.701657$$

$$\approx 12.7m$$

15.

a) $x = 10 + 8t - 2t^2$
 i) $x = 0 \quad 2t^2 - 8t - 10 = 0$
 $t^2 - 4t - 5 = 0$
 $(t - 5)(t + 1) = 0$
 $t = 5, -1$

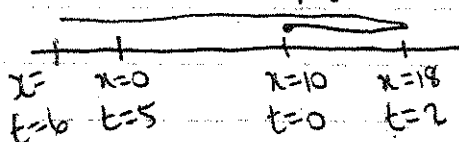
$t \geq 0 \therefore t = 5 \text{ seconds} \checkmark$

ii) $\dot{x} = 8 - 4t = 0$

$10 + 8 \times 2 - 2 \times 2^2 = 18 \checkmark$
 $4t = 8 \quad t = 2 \text{ seconds} \checkmark$

iii) $\ddot{x} = -4 = \text{constant} \checkmark$

iv) $t = 2 \quad x = 10 + 8 \times 2 - 2 \times 2^2 = 18$



$a = 10 + 8 \times 6 - 2 \times 6^2 = -14 \checkmark$

$d = 0 + 18 + 14 = 40 \text{ m} \checkmark$

b) $y = \frac{2}{1+x^2} \quad V = \pi \int_a^b x^2 dy$

$1+x^2 = \frac{2}{y}$

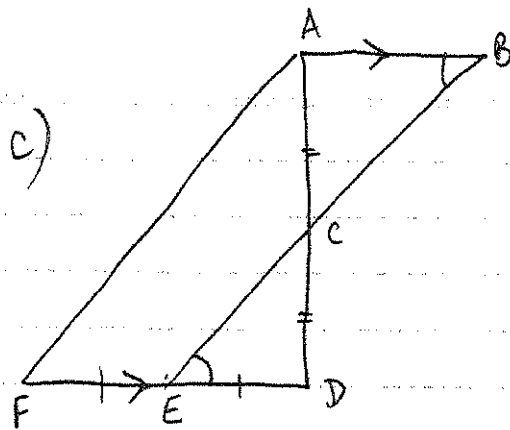
$x^2 = \frac{2}{y} - 1 \checkmark$

$V = \pi \int_1^2 \left(\frac{2}{y} - 1 \right) dy$

i) $V = \pi \left[2 \ln y - y \right]_1^2 \checkmark$

$= \pi \left[(2 \ln 2 - 2) - (2 \ln 1 - 1) \right]$

$= 2\pi \ln 2 - \pi \quad u^3 \checkmark$



i) Alternate angles are equal
 $AB \parallel FD$

ii) In $\triangle CDE$ & $\triangle CAB$

$AC = CD$ given

$\angle ECD = \angle ACB$ vertically opposite

$\angle DEC = \angle ABC$ Alternate angles equal in parallel lines

$\therefore \triangle CDE \equiv \triangle CAB$ AAS

iii) $CE = BC$ corresponding sides in congruent triangles

$\triangle CED \sim \triangle AFD$

$\frac{AF}{CE} = \frac{AD}{CD} = \frac{2}{1}$

$\therefore AF = 2 \times CE$

So $AF = 2BC$

d) $f(x) = \ln(x^2 + 1)$

$\int_0^4 \log_e(x^2 + 1) dx \approx \frac{1}{3} \left[f(0) + f(4) + 4(f(1) + f(3)) + 2f(2) \right]$

$= \frac{1}{3} \left[\ln 1 + \ln 17 + 4(\ln 2 + \ln 10) + 2 \ln 5 \right]$

$= \frac{1}{3} \left[\ln 17 + \ln 20^4 + \ln 5^2 \right]$

$= \frac{1}{3} \ln(17 \times 20^4 \times 5^2) \approx 6.01167$

$= \frac{1}{3} \ln 68000000$

$$16a) P = 2000 e^{kt}$$

$$i) \frac{dP}{dt} = 2000 \times k e^{kt}$$

$$= k \times 2000 e^{kt}$$

$$\frac{dP}{dt} = kP \quad \checkmark$$

$$ii) P = 2P_0 \text{ when } t = 4$$

$$4000 = 2000 e^{4k}$$

$$2 = e^{4k} \quad \checkmark$$

$$\ln 2 = 4k$$

$$k = \frac{1}{4} \ln 2 \quad \checkmark$$

$$\approx 0.173$$

$$iii) t = 6 \quad 0.173 \times 6 \quad \checkmark$$

$$P = 2000 \times e$$

$$= 5656.85$$

$$\approx 5657 \text{ (if exact } k)$$

$$\text{or } P \approx 5647 \text{ for rounded } k.$$

$$b) r = 7.2\% = \frac{0.072}{12} = 0.006$$

$$A_1 = 200000 + 0.006 \times 200000 - M$$

$$= 200000 \times 1.006 - M$$

$$A_2 = A_1 \times 1.006 - M \quad \checkmark$$

$$= 200000 \times 1.006^2 - M \times 1.006 - M$$

$$A_n = 200000 \times 1.006^n - M \times 1.006^{n-1} \dots$$

$$- M \times 1.006 - M \quad \checkmark$$

$$A_n = 200000 \times 1.006^n - M \left(1 + 1.006 + 1.006^2 + \dots + 1.006^{n-1} \right)$$

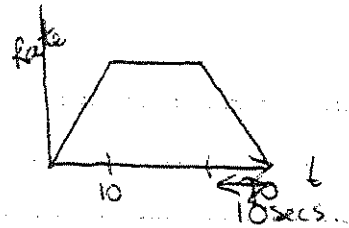
$$n = 25 \times 12 = 300$$

$$ii) 0 = 200000 \times 1.006^{300} - M \left[\frac{1(1.006^{300} - 1)}{1.006 - 1} \right] \quad \checkmark$$

$$M = \frac{200000 \times 1.006^{300} \times 0.006}{(1.006^{300} - 1)}$$

$$= \$1439.18 \quad \checkmark$$

$$c) R = \frac{6t}{50}$$



$$\frac{dV}{dt} = \frac{6t}{50}$$

$$V = \frac{6}{50} \times \frac{t^2}{2} + C$$

$$V = \frac{3t^2}{50} + C \quad \text{Initially } V=0 \text{ at } t=0$$

$$0 = \frac{3 \times 0^2}{50} + C \quad C=0$$

$$V = \frac{3t^2}{50} \quad \text{for } 0 \leq t \leq 10 \quad \checkmark$$

$$\text{At } t=10 \quad V = \frac{3 \times 10^2}{50} = 6L$$

$$\text{At } t=10 \quad R = \frac{6 \times 10}{50} = \frac{6}{5} L/s$$

$$\therefore \frac{dV}{dt} = \frac{6}{5} \quad \checkmark \quad \text{In this section}$$

$$V = \frac{6}{5} t + C \quad t=10 \quad V=6$$

$$6 = \frac{6}{5} \times 10 + C \quad C=-6$$

$$V = \frac{6}{5} t - 6$$

$$= \frac{6}{5} (t - 5) \quad \checkmark \quad \text{for } t \geq 10$$

$$ii) \text{ First 10 seconds Volume} = 6L$$

$$\text{last 10 seconds } R = \frac{6}{5} L/s$$

$$\therefore V = \frac{6}{5} \times 10 = 12L \quad \checkmark$$

$$\therefore 120 - (6 + 12) = 102$$

$$102 = \frac{6}{5} (t - 5)$$

$$\frac{102 \times 5}{6} + 5 = t \quad t = 90 \text{ seconds} \quad \checkmark$$